Projeto: Predição de Inadimplência de Cartão de Crédito

Machine Learning Supervisionado com Scikit-Learn

Dataset: Default of Credit Card Clients (UCI)

Objetivo: Construir, avaliar e interpretar modelos supervisionados clássicos para predição de default (inadimplência) no mês seguinte.

---

1. Instalação e Imports

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import time
from sklearn.model_selection import train_test_split, GridSearchCV, StratifiedKFold, RandomizedSearchCV
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer, KNNImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn import tree, set_config
from sklearn.linear_model import Perceptron
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, confusion_matrix, ConfusionMatrixDisplay
)
from sklearn.model_selection import GridSearchCV, StratifiedKFold


import warnings
warnings.filterwarnings('ignore')

sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (10, 6)

print('Bibliotecas importadas com sucesso!')
```

```
Bibliotecas importadas com sucesso!
```

2. Upload e Carregamento do Dataset

```
from google.colab import files
print('Aguardando upload do arquivo...')
uploaded = files.upload()
```

```
Aguardando upload do arquivo...
```

Escolher arquivos default of cr…rd clients.xls

default of credit card clients.xls(application/vnd.ms-excel) - 5539328 bytes, last modified: 22/05/2023 - 100% done

```
Saving default of credit card clients.xls to default of credit card clients.xls
```

```
import os
filename = list(uploaded.keys())[0]
print(f' Arquivo carregado: {filename}')

if filename.endswith('.csv'):
    df = pd.read_csv(filename)
elif filename.endswith('.xls') or filename.endswith('.xlsx'):
    df = pd.read_excel(filename, header=1)
else:
    print(' Formato não reconhecido. Use .csv ou .xls')

print(' Dataset carregado!')
print(f' Shape: {df.shape}')
df.head()
```

```
Arquivo carregado: default of credit card clients.xls
Dataset carregado!
Shape: (30000, 25)
```

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	...	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	P
0	1	20000	2	2	1	24	2	2	-1	-1	...	0	0	0	0	
1	2	120000	2	2	2	26	-1	2	0	0	...	3272	3455	3261	0	
2	3	90000	2	2	2	34	0	0	0	0	...	14331	14948	15549	1518	
3	4	50000	2	2	1	37	0	0	0	0	...	28314	28959	29547	2000	

3. Análise Exploratória Inicial

```
print('ℹ Informações gerais do dataset:')
print(df.info())
print('\nℹ Estatísticas descritivas:')
print(df.describe())
```

```
 23  PAY_AMT6                    30000 non-null  int64
 24  default payment next month  30000 non-null  int64
dtypes: int64(25)
memory usage: 5.7 MB
None

ℹ Estatísticas descritivas:
                 ID       LIMIT_BAL           SEX     EDUCATION      MARRIAGE  \
count  30000.000000    30000.000000  30000.000000  30000.000000  30000.000000
mean   15000.500000   167484.322667      1.603733      1.853133      1.551867
std     8660.398374   129747.661567      0.489129      0.790349      0.521970
min        1.000000    10000.000000      1.000000      0.000000      0.000000
25%     7500.750000    50000.000000      1.000000      1.000000      1.000000
50%    15000.500000   140000.000000      2.000000      2.000000      2.000000
75%    22500.250000   240000.000000      2.000000      2.000000      2.000000
max    30000.000000  1000000.000000      2.000000      6.000000      3.000000

               AGE         PAY_0         PAY_2         PAY_3         PAY_4  \
count  30000.000000  30000.000000  30000.000000  30000.000000  30000.000000
mean      35.485500     -0.016700     -0.133767     -0.166200     -0.220667
std        9.217904      1.123802      1.197186      1.196868      1.169139
min       21.000000     -2.000000     -2.000000     -2.000000     -2.000000
25%       28.000000     -1.000000     -1.000000     -1.000000     -1.000000
50%       34.000000      0.000000      0.000000      0.000000      0.000000
75%       41.000000      0.000000      0.000000      0.000000      0.000000
max       79.000000      8.000000      8.000000      8.000000      8.000000

       ...      BILL_AMT4      BILL_AMT5      BILL_AMT6       PAY_AMT1  \
count  ...   30000.000000   30000.000000   30000.000000   30000.000000
mean   ...   43262.948967   40311.400967   38871.760400    5663.580500
std    ...   64332.856134   60797.155770   59554.107537   16563.280354
min    ... -170000.000000  -81334.000000 -339603.000000       0.000000
25%    ...    2326.750000    1763.000000    1256.000000    1000.000000
50%    ...   19052.000000   18104.500000   17071.000000    2100.000000
75%    ...   54506.000000   50190.500000   49198.250000    5006.000000
max    ...  891586.000000  927171.000000  961664.000000  873552.000000

           PAY_AMT2      PAY_AMT3       PAY_AMT4       PAY_AMT5  \
count  3.000000e+04   30000.00000   30000.000000   30000.000000
mean   5.921163e+03    5225.68150    4826.076867    4799.387633
std    2.304087e+04   17606.96147   15666.159744   15278.305679
min    0.000000e+00       0.00000       0.000000       0.000000
25%    8.330000e+02     390.00000     296.000000     252.500000
50%    2.009000e+03    1800.00000    1500.000000    1500.000000
75%    5.000000e+03    4505.00000    4013.250000    4031.500000
max    1.684259e+06  896040.00000  621000.000000  426529.000000

            PAY_AMT6  default payment next month
count   30000.000000                30000.000000
mean     5215.502567                    0.221200
std     17777.465775                    0.415062
min         0.000000                    0.000000
25%       117.750000                    0.000000
50%      1500.000000                    0.000000
75%      4000.000000                    0.000000
max    528666.000000                    1.000000

[8 rows x 25 columns]
```

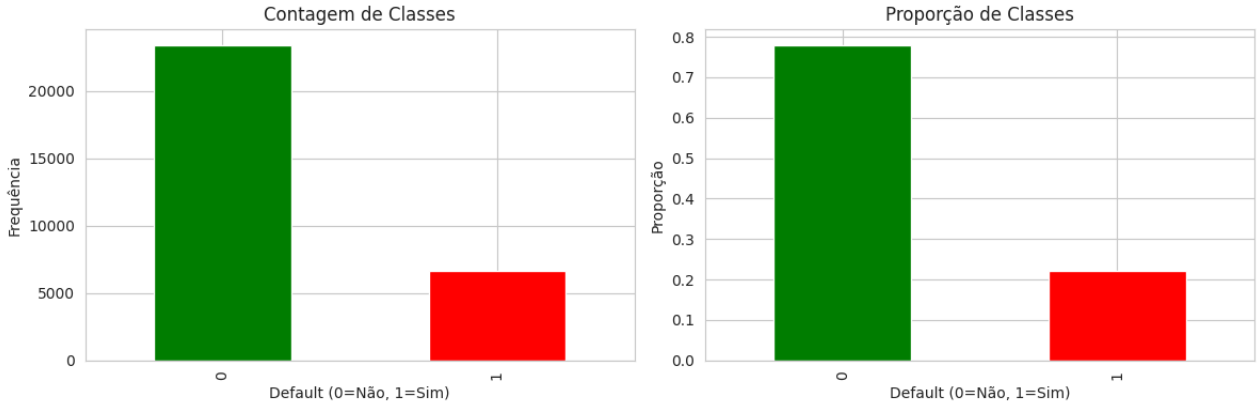
```
print('ℹColunas disponíveis:')
print(df.columns.tolist())
```

```
if 'default.payment.next.month' in df.columns:
    df = df.rename(columns={'default.payment.next.month': 'default'})
elif 'default payment next month' in df.columns:
    df = df.rename(columns={'default payment next month': 'default'})

target_col = 'default'
print(f'\n Coluna alvo definida: {target_col}')
```

```
Colunas disponíveis:
['ID', 'LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1'

 Coluna alvo definida: default
```

```
print('Distribuição de Classes:')
print(df[target_col].value_counts())
print('\nProporção:')
print(df[target_col].value_counts(normalize=True))

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
df[target_col].value_counts().plot(kind='bar', ax=ax[0], color=['green', 'red'])
ax[0].set_title('Contagem de Classes')
ax[0].set_xlabel('Default (0=Não, 1=Sim)')
ax[0].set_ylabel('Frequência')

df[target_col].value_counts(normalize=True).plot(kind='bar', ax=ax[1], color=['green', 'red'])
ax[1].set_title('Proporção de Classes')
ax[1].set_xlabel('Default (0=Não, 1=Sim)')
ax[1].set_ylabel('Proporção')
plt.tight_layout()
plt.show()
```

```
Distribuição de Classes:
default
0    23364
1     6636
Name: count, dtype: int64

Proporção:
default
0    0.7788
1    0.2212
Name: proportion, dtype: float64
```



Observações sobre desbalanceamento

O dataset apresenta **desbalanceamento de classes**, com a maioria dos clientes não entrando em default. Isso exige atenção especial às métricas **Precision, Recall e F1-score**.

4. Preparação dos Dados

```
X = df.drop(columns=[target_col])
y = df[target_col]

categorical_cols = [col for col in ['SEX', 'EDUCATION', 'MARRIAGE'] if col in X.columns]
numeric_cols = [c for c in X.columns if c not in categorical_cols]

print(f'Features separadas!')
```

```
print(f'Categóricas ({len(categorical_cols)}): {categorical_cols}')
print(f' Numéricas ({len(numeric_cols)}): {numeric_cols}')
```

```
Features separadas!
Categóricas (3): ['SEX', 'EDUCATION', 'MARRIAGE']
 Numéricas (21): ['ID', 'LIMIT_BAL', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL_AMT2',
```

```
total_ausentes = X.isnull().sum().sum()
if total_ausentes == 0:
    print(' Nenhum valor ausente encontrado!')
else:
    print(f' Total de valores ausentes: {total_ausentes}')
    print(X.isnull().sum()[X.isnull().sum() > 0])
```

```
 Nenhum valor ausente encontrado!
```

5. Split Treino/Teste

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print(' Dados divididos com sucesso!')
print(f' Treino: {X_train.shape[0]} amostras')
print(f' Teste:  {X_test.shape[0]} amostras')
print(f'\nDistribuição no treino:')
print(y_train.value_counts(normalize=True))
print(f'\nDistribuição no teste:')
print(y_test.value_counts(normalize=True))
```

```
 Dados divididos com sucesso!
 Treino: 24000 amostras
 Teste:  6000 amostras

Distribuição no treino:
default
0    0.778792
1    0.221208
Name: proportion, dtype: float64

Distribuição no teste:
default
0    0.778833
1    0.221167
Name: proportion, dtype: float64
```

6. Pipeline de Pré-processamento

O pipeline abaixo replica exatamente o diagrama do projeto:

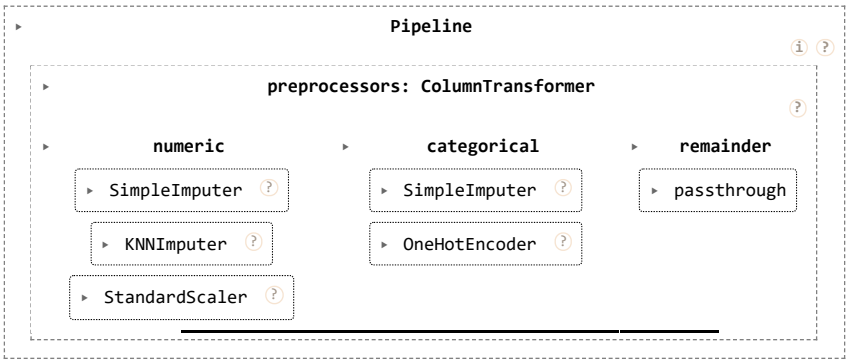
- **SimpleImputer**: preenche valores ausentes
- **KNNImputer**: imputação por vizinhos mais próximos (numéricas)
- **OneHotEncoder**: encoding de categóricas
- **StandardScaler**: padronização de numéricas
- **remainder passthrough**: mantém colunas sem transformação

```
# Ativa visualização do diagrama
set_config(display='diagram')

# Pipeline para colunas NUMÉRICAS
numeric_pipeline = Pipeline(steps=[
    ('simple_imputer', SimpleImputer(strategy='median')),
    ('knn_imputer',    KNNImputer(n_neighbors=5)),
    ('standardscale',  StandardScaler())
])

# Pipeline para colunas CATEGÓRICAS
categorical_pipeline = Pipeline(steps=[
    ('simple_imputer',  SimpleImputer(strategy='most_frequent')),
    ('onehotencoder',   OneHotEncoder(drop='first', handle_unknown='ignore'))
])

# ColumnTransformer
preprocessor = ColumnTransformer(
```

```
    transformers=[
        ('numeric',      numeric_pipeline,      numeric_cols),
        ('categorical',  categorical_pipeline,  categorical_cols)
    ],
    remainder='passthrough'
)

print(' Preprocessor criado com sucesso!')
```

```
 Preprocessor criado com sucesso!
```

```
#  diagrama do pipeline
meu_projeto = Pipeline(steps=[
    ('preprocessors', preprocessor)
])

meu_projeto
```



7. Funções Auxiliares de Avaliação

```
def print_metrics(y_true, y_pred, title='Modelo'):
    print(f"\n{'='*60}")
    print(f"{title:^60}")
    print('='*60)
    print(f"Accuracy : {accuracy_score(y_true, y_pred):.4f}")
    print(f"Precision: {precision_score(y_true, y_pred, zero_division=0):.4f}")
    print(f"Recall   : {recall_score(y_true, y_pred, zero_division=0):.4f}")
    print(f"F1-score : {f1_score(y_true, y_pred, zero_division=0):.4f}")
    print('='*60)

def plot_confusion_matrix(y_true, y_pred, title='Confusion Matrix'):
    cm = confusion_matrix(y_true, y_pred)
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=['Não Default', 'Default'])
    disp.plot(cmap='Blues')
    plt.title(title)
    plt.show()

print('Funções auxiliares definidas!')
```

```
Funções auxiliares definidas!
```

PARTE 1: MODELO BASELINE - PERCEPTRON

O **Perceptron** é um classificador linear clássico que aprende um hiperplano separador no espaço das features. Serve como **baseline** porque:

- Tem baixa complexidade computacional
- Fornece coeficientes lineares explícitos
- Estabelece ponto de comparação para modelos mais sofisticados

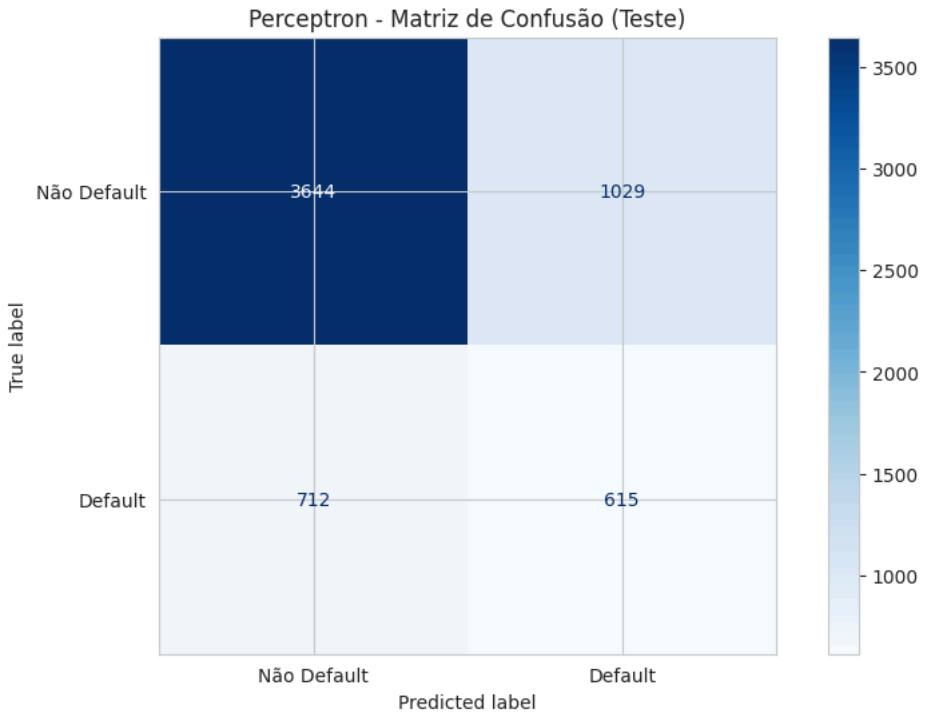
```
print(' Treinando modelo Perceptron (baseline)...')

perceptron_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier',   Perceptron(random_state=42, max_iter=1000))
])

perceptron_pipeline.fit(X_train, y_train)
```

```
y_pred_train_perc = perceptron_pipeline.predict(X_train)
y_pred_test_perc  = perceptron_pipeline.predict(X_test)

print(' Perceptron treinado!')
```

```
 Treinando modelo Perceptron (baseline)...
 Perceptron treinado!
```

```
print(' Avaliando Perceptron...')
print_metrics(y_train, y_pred_train_perc, 'PERCEPTRON - TREINO')
print_metrics(y_test,  y_pred_test_perc,  'PERCEPTRON - TESTE')

print('\n=== Classification Report (Teste) ===')
print(classification_report(y_test, y_pred_test_perc,
                            target_names=['Não Default', 'Default'],
                            zero_division=0))

plot_confusion_matrix(y_test, y_pred_test_perc, 'Perceptron - Matriz de Confusão (Teste)')
```

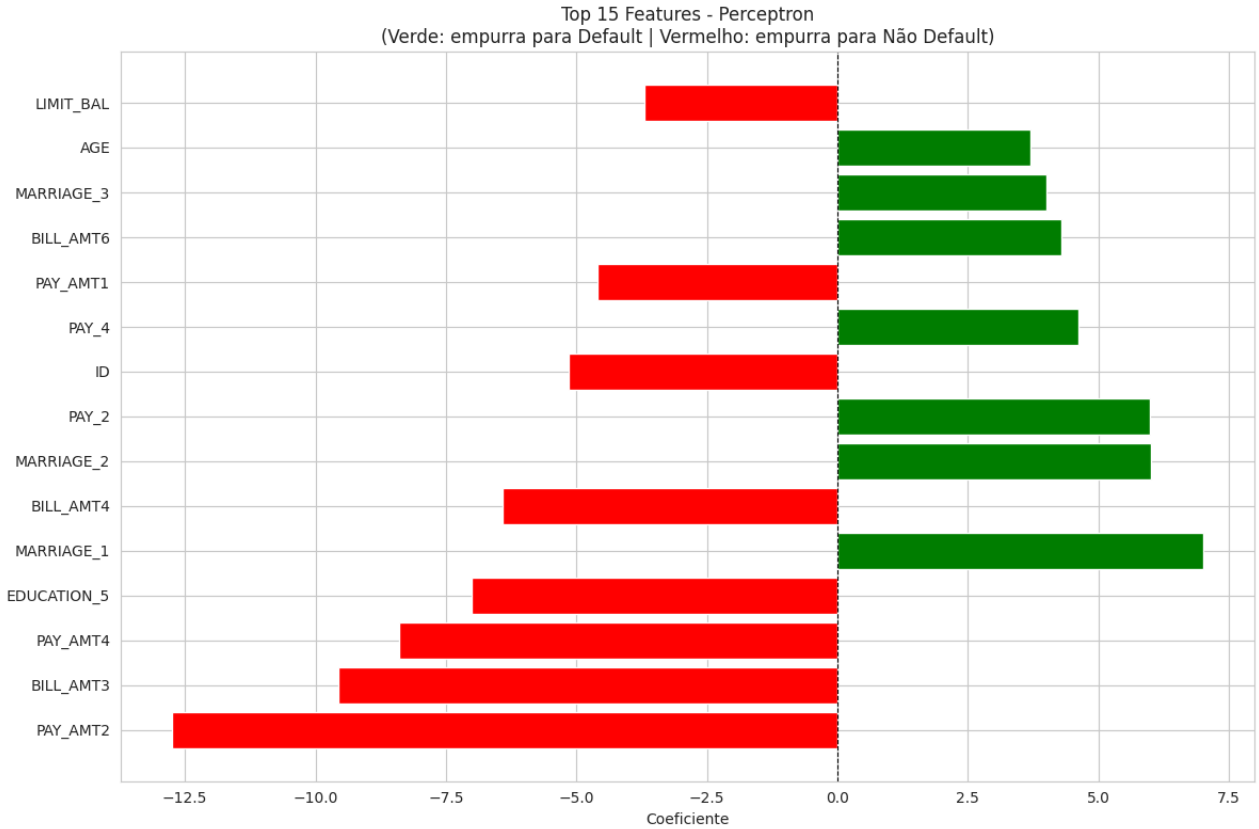
```
 Avaliando Perceptron...

============================================================
                     PERCEPTRON - TREINO
============================================================
Accuracy : 0.7215
Precision: 0.3964
Recall   : 0.4954
F1-score : 0.4404
============================================================

============================================================
                      PERCEPTRON - TESTE
============================================================
Accuracy : 0.7098
Precision: 0.3741
Recall   : 0.4635
F1-score : 0.4140
============================================================

=== Classification Report (Teste) ===
              precision    recall  f1-score   support

 Não Default       0.84      0.78      0.81      4673
     Default       0.37      0.46      0.41      1327

    accuracy                           0.71      6000
   macro avg       0.61      0.62      0.61      6000
weighted avg       0.73      0.71      0.72      6000
```




```
print('ℹ Extraindo coeficientes do Perceptron...')

perceptron_model     = perceptron_pipeline.named_steps['classifier']
preprocessor_fitted  = perceptron_pipeline.named_steps['preprocessor']
```

```
feature_names_transformed = list(numeric_cols)

if categorical_cols:
    cat_encoder = preprocessor_fitted.named_transformers_['categorical'].named_steps['onehotencoder']
    cat_feature_names = cat_encoder.get_feature_names_out(categorical_cols)
    feature_names_transformed.extend(cat_feature_names)

coef = perceptron_model.coef_[0]
bias = perceptron_model.intercept_[0]

coef_df = pd.DataFrame({
    'Feature':     feature_names_transformed[:len(coef)],
    'Coeficiente': coef
}).sort_values(by='Coeficiente', key=abs, ascending=False)

print(f'\nBias (w0): {bias:.4f}')
print('\nTop 10 Features mais influentes:')
print(coef_df.head(10).to_string(index=False))

plt.figure(figsize=(12, 8))
top_features = coef_df.head(15)
colors = ['red' if x < 0 else 'green' for x in top_features['Coeficiente']]
plt.barh(top_features['Feature'], top_features['Coeficiente'], color=colors)
plt.xlabel('Coeficiente')
plt.title('Top 15 Features - Perceptron\n(Verde: empurra para Default | Vermelho: empurra para Não Default)')
plt.axvline(0, color='black', linestyle='--', linewidth=0.8)
plt.tight_layout()
plt.show()

print(' Interpretação do Perceptron concluída!')
```

```
ℹ Extraindo coeficientes do Perceptron...

Bias (w0): -21.0000

Top 10 Features mais influentes:
    Feature  Coeficiente
   PAY_AMT2   -12.742513
  BILL_AMT3    -9.554759
   PAY_AMT4    -8.402845
EDUCATION_5    -7.000000
 MARRIAGE_1     7.000000
  BILL_AMT4    -6.421884
 MARRIAGE_2     6.000000
      PAY_2     5.992902
         ID    -5.158332
      PAY_4     4.606248
```




```
Interpretação do Perceptron concluída!
```

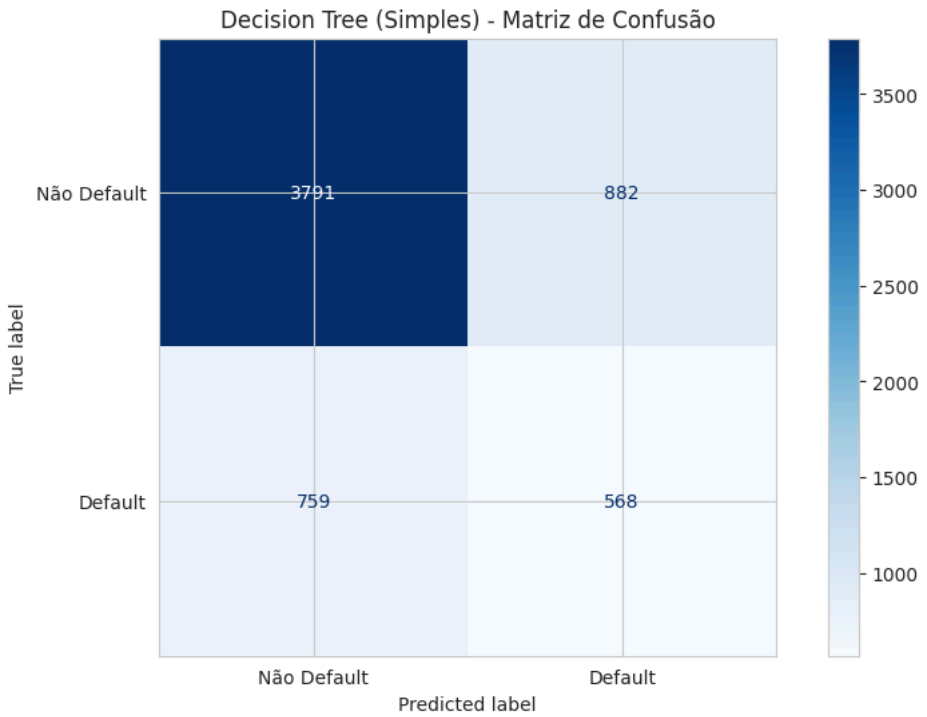
PARTE 2: ÁRVORE DE DECISÃO

```
print(' Treinando Árvore de Decisão (simples)...')

dt_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier',   DecisionTreeClassifier(random_state=42))
])

dt_pipeline.fit(X_train, y_train)

y_pred_train_dt = dt_pipeline.predict(X_train)
y_pred_test_dt  = dt_pipeline.predict(X_test)

print(' Árvore de Decisão (simples) treinada!')
```

```
Treinando Árvore de Decisão (simples)...
Árvore de Decisão (simples) treinada!
```

```
print(' Avaliando Árvore de Decisão (simples)...')
print_metrics(y_train, y_pred_train_dt, 'DECISION TREE (SIMPLES) - TREINO')
print_metrics(y_test,  y_pred_test_dt,  'DECISION TREE (SIMPLES) - TESTE')

print('\n=== Classification Report (Teste) ===')
print(classification_report(y_test, y_pred_test_dt,
                            target_names=['Não Default', 'Default'],
                            zero_division=0))

plot_confusion_matrix(y_test, y_pred_test_dt, 'Decision Tree (Simples) - Matriz de Confusão')
```

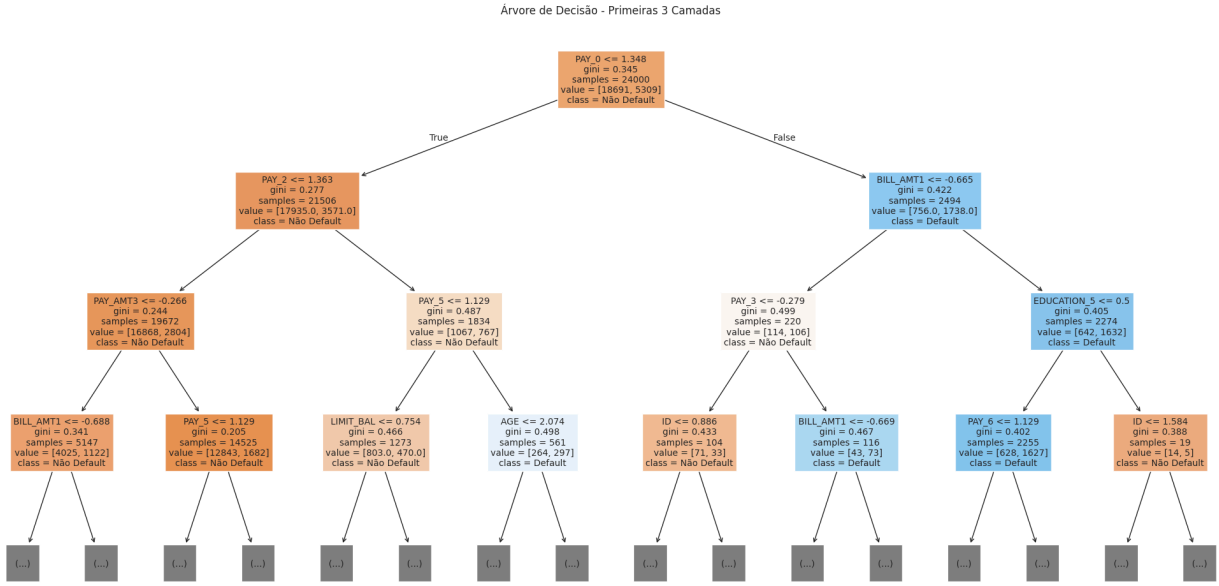
```
 Avaliando Árvore de Decisão (simples)...

============================================================
              DECISION TREE (SIMPLES) - TREINO
============================================================
Accuracy : 1.0000
Precision: 1.0000
Recall   : 1.0000
F1-score : 1.0000
============================================================

============================================================
              DECISION TREE (SIMPLES) - TESTE
============================================================
Accuracy : 0.7265
Precision: 0.3917
Recall   : 0.4280
F1-score : 0.4091
============================================================

=== Classification Report (Teste) ===
              precision    recall  f1-score   support

 Não Default       0.83      0.81      0.82      4673
     Default       0.39      0.43      0.41      1327

    accuracy                           0.73      6000
   macro avg       0.61      0.62      0.62      6000
weighted avg       0.74      0.73      0.73      6000
```




```
print('Gerando visualização da Árvore de Decisão...')

dt_model = dt_pipeline.named_steps['classifier']

# Nomes das features após preprocessamento
feature_names_tree = list(numeric_cols)
if categorical_cols:
    cat_enc = dt_pipeline.named_steps['preprocessor'].named_transformers_['categorical'].named_steps['onehotencoder']
    feature_names_tree.extend(cat_enc.get_feature_names_out(categorical_cols))

plt.figure(figsize=(20, 10))
tree.plot_tree(dt_model,
```

```
                feature_names=feature_names_tree,
                class_names=['Não Default', 'Default'],
                filled=True,
                max_depth=3,
                fontsize=10)
plt.title('Árvore de Decisão - Primeiras 3 Camadas')
plt.tight_layout()
plt.show()

print(f' Profundidade da árvore: {dt_model.get_depth()}')
print(f' Número de folhas: {dt_model.get_n_leaves()}')
```

```
Gerando visualização da Árvore de Decisão...
```




```
Profundidade da árvore: 43
Número de folhas: 3544
```

PARTE 3: VALIDAÇÃO CRUZADA E OTIMIZAÇÃO

```
# Grid reduzido para ser mais rápido (recomendado)
param_grid_dt = {
    'classifier__max_depth':         [3, 5, 10, None],
    'classifier__min_samples_split': [2, 10, 20],
    'classifier__min_samples_leaf':  [1, 5, 10],
```

```
    'classifier__criterion':         ['gini', 'entropy']
}

total_combinacoes = np.prod([len(v) for v in param_grid_dt.values()])
n_folds = 5
total_fits = total_combinacoes * n_folds

print(f'Combinações de hiperparâmetros : {total_combinacoes}')
print(f'Folds de validação cruzada     : {n_folds}')
print(f'Total de fits a executar       : {total_fits}')
print('='*60)

cv_strategy = StratifiedKFold(n_splits=n_folds, shuffle=True, random_state=42)

dt_pipeline_base = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier',   DecisionTreeClassifier(random_state=42))
])

# verbose=3 mostra cada fit individualmente
grid_search_dt = GridSearchCV(
    estimator=dt_pipeline_base,
    param_grid=param_grid_dt,
    scoring='f1',
    cv=cv_strategy,
    n_jobs=1,         # <-- coloca 1 para os prints aparecerem em ordem
    verbose=3,        # <-- mostra cada fold + candidato
    return_train_score=True
)

print('Iniciando Grid Search para Árvore de Decisão...')
print('='*60)

inicio = time.time()
grid_search_dt.fit(X_train, y_train)
fim = time.time()

tempo_total = fim - inicio
print('='*60)
print(f' Grid Search concluído!')
print(f' Tempo total: {tempo_total:.1f} segundos ({tempo_total/60:.1f} minutos)')
print(f' Melhor F1-Score (CV): {grid_search_dt.best_score_:.4f}')
print(f' Melhores parâmetros:')
for param, value in grid_search_dt.best_params_.items():
    print(f'   {param}: {value}')
```

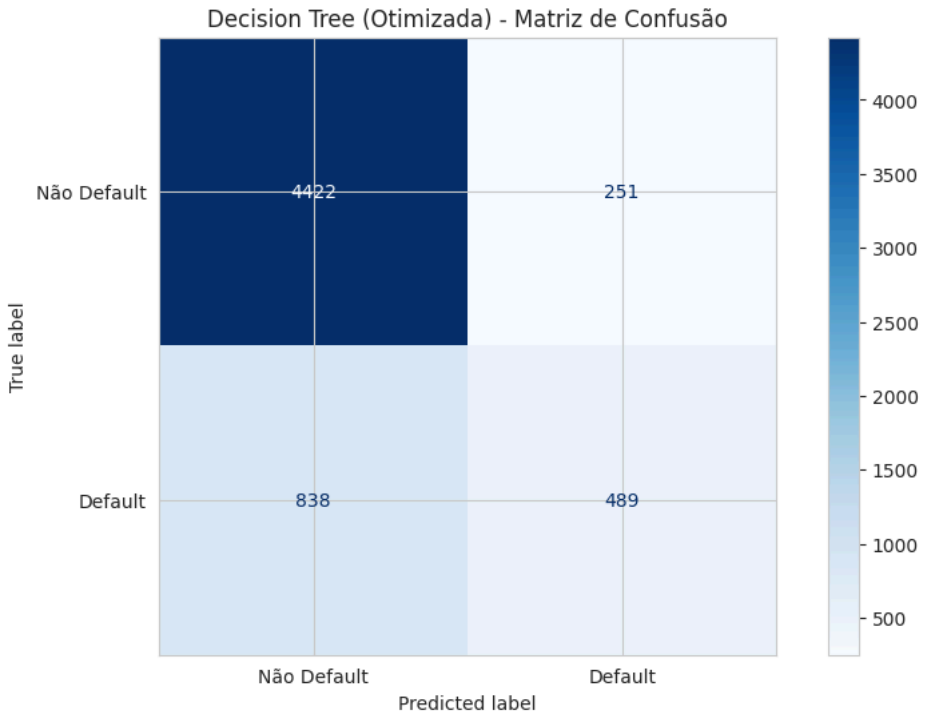
```
[CV 3/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 4/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 5/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 1/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 2/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 3/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 4/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
[CV 5/5] END classifier__criterion=entropy, classifier__max_depth=None, classifier__min_samples_leaf=10, classifier__min_s
============================================================
 Grid Search concluído!
 Tempo total: 231.5 segundos (3.9 minutos)
 Melhor F1-Score (CV): 0.4846
 Melhores parâmetros:
   classifier__criterion: gini
   classifier__max_depth: 3
   classifier__min_samples_leaf: 1
   classifier__min_samples_split: 2
```

```
print('\n' + '='*80)
print('RESULTADOS DO GRID SEARCH')
print('='*80)
print('\n Melhores hiperparâmetros:')
for param, value in grid_search_dt.best_params_.items():
    print(f'  {param}: {value}')
print(f'\n Melhor F1-Score médio (CV): {grid_search_dt.best_score_:.4f}')

best_dt = grid_search_dt.best_estimator_
y_pred_train_dt_opt = best_dt.predict(X_train)
y_pred_test_dt_opt  = best_dt.predict(X_test)
```

```
================================================================================
RESULTADOS DO GRID SEARCH
================================================================================

 Melhores hiperparâmetros:
  classifier__criterion: gini
  classifier__max_depth: 3
  classifier__min_samples_leaf: 1
  classifier__min_samples_split: 2

 Melhor F1-Score médio (CV): 0.4846
```

```
print('Avaliando Árvore Otimizada...')
print_metrics(y_train, y_pred_train_dt_opt, 'DECISION TREE (OTIMIZADA) - TREINO')
print_metrics(y_test,  y_pred_test_dt_opt,  'DECISION TREE (OTIMIZADA) - TESTE')

print('\n=== Classification Report (Teste) ===')
print(classification_report(y_test, y_pred_test_dt_opt,
                            target_names=['Não Default', 'Default'],
                            zero_division=0))

plot_confusion_matrix(y_test, y_pred_test_dt_opt, 'Decision Tree (Otimizada) - Matriz de Confusão')
```

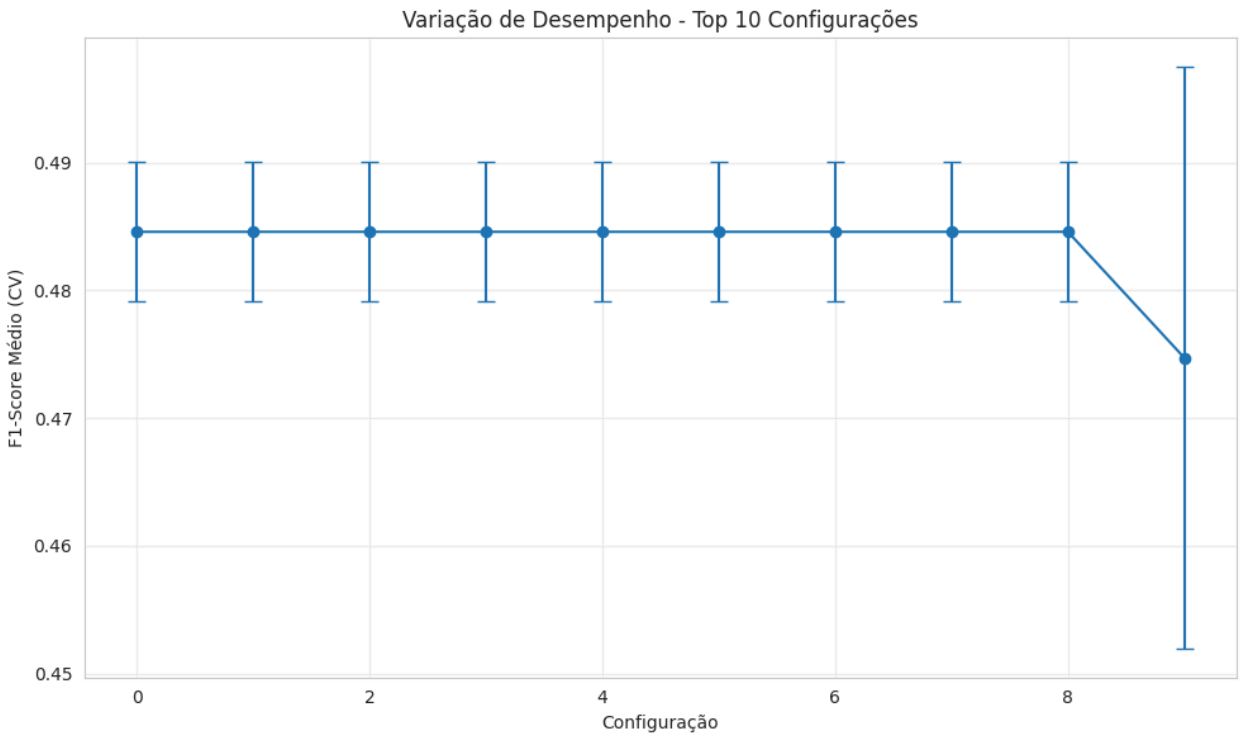
```
Avaliando Árvore Otimizada...

============================================================
             DECISION TREE (OTIMIZADA) - TREINO
============================================================
Accuracy : 0.8230
Precision: 0.6811
Recall   : 0.3762
F1-score : 0.4846
============================================================

============================================================
             DECISION TREE (OTIMIZADA) - TESTE
============================================================
Accuracy : 0.8185
Precision: 0.6608
Recall   : 0.3685
F1-score : 0.4731
============================================================

=== Classification Report (Teste) ===
              precision    recall  f1-score   support

 Não Default       0.84      0.95      0.89      4673
     Default       0.66      0.37      0.47      1327

    accuracy                           0.82      6000
   macro avg       0.75      0.66      0.68      6000
weighted avg       0.80      0.82      0.80      6000
```




```
print(' Analisando robustez (variação entre folds)...')

cv_results  = pd.DataFrame(grid_search_dt.cv_results_)
top_configs = cv_results.nsmallest(10, 'rank_test_score')[[
    'params', 'mean_test_score', 'std_test_score', 'mean_train_score'
]]

print('\nTop 10 configurações:')
for _, row in top_configs.iterrows():
    print(f"  Params: {row['params']}")
    print(f"  Test:  {row['mean_test_score']:.4f} (+/- {row['std_test_score']:.4f})")
    print(f"  Train: {row['mean_train_score']:.4f}\n")

plt.figure(figsize=(10, 6))
plt.errorbar(range(len(top_configs)),
             top_configs['mean_test_score'],
             yerr=top_configs['std_test_score'],
             fmt='o-', capsize=5)
plt.xlabel('Configuração')
plt.ylabel('F1-Score Médio (CV)')
plt.title('Variação de Desempenho - Top 10 Configurações')
plt.grid(True, alpha=0.3)
plt.tight_layout()
```

```
plt.show()

print(' Análise de robustez concluída!')
```

```
 Analisando robustez (variação entre folds)...

Top 10 configurações:
  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 1, 'classifier__mi
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 1, 'classifier__mi
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 1, 'classifier__mi
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 5, 'classifier__mi
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 5, 'classifier__mi
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 5, 'classifier__mi
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 10, 'classifier__m
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 10, 'classifier__m
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'gini', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 10, 'classifier__m
  Test:  0.4846 (+/- 0.0055)
  Train: 0.4852

  Params: {'classifier__criterion': 'entropy', 'classifier__max_depth': 3, 'classifier__min_samples_leaf': 1, 'classifier_
  Test:  0.4747 (+/- 0.0228)
  Train: 0.4736
```




```
 Análise de robustez concluída!
```

PARTE 4: RANDOM FOREST

```
param_dist_rf = {
    'classifier__n_estimators':      [100, 200, 300],
    'classifier__max_depth':         [5, 10, 15, 20, None],
    'classifier__min_samples_split': [2, 10, 20],
    'classifier__min_samples_leaf':  [1, 2, 5, 10],
    'classifier__max_features':      ['sqrt', 'log2', 0.5]
}

n_iter   = 10
n_folds  = 5
total_fits = n_iter * n_folds

print(f'Combinações sorteadas (n_iter) : {n_iter}')
print(f'Folds de validação cruzada     : {n_folds}')
print(f'Total de fits a executar       : {total_fits}')
print('='*60)

rf_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier',   RandomForestClassifier(random_state=42, n_jobs=1))
])

random_search_rf = RandomizedSearchCV(
    estimator=rf_pipeline,
    param_distributions=param_dist_rf,
    n_iter=n_iter,
    scoring='f1',
    cv=cv_strategy,
    n_jobs=1,          # <-- 1 para prints aparecerem em ordem
    random_state=42,
    verbose=3,         # <-- mostra cada fold + candidato
    return_train_score=True
)

print('⏳ Iniciando Random Search para Random Forest...')
print('='*60)

inicio = time.time()
random_search_rf.fit(X_train, y_train)
fim = time.time()

tempo_total = fim - inicio

print('='*60)
print(f'Random Forest otimizado com sucesso!')
print(f'Tempo total: {tempo_total:.1f} segundos ({tempo_total/60:.1f} minutos)')
print(f'Melhor F1-Score (CV): {random_search_rf.best_score_:.4f}')
print(f'Melhores parâmetros:')
for param, value in random_search_rf.best_params_.items():
    print(f'   {param}: {value}')

best_rf         = random_search_rf.best_estimator_
y_pred_train_rf = best_rf.predict(X_train)
y_pred_test_rf  = best_rf.predict(X_test)
```

```
[CV 4/5] END classifier__max_depth=15, classifier__max_features=sqrt, classifier__min_samples_leaf=2, classifier__min_samp
[CV 5/5] END classifier__max_depth=15, classifier__max_features=sqrt, classifier__min_samples_leaf=2, classifier__min_samp
[CV 1/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=1, classifier__min_sample
[CV 2/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=1, classifier__min_sample
[CV 3/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=1, classifier__min_sample
[CV 4/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=1, classifier__min_sample
[CV 5/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=1, classifier__min_sample
[CV 1/5] END classifier__max_depth=None, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sam
[CV 2/5] END classifier__max_depth=None, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sam
[CV 3/5] END classifier__max_depth=None, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sam
[CV 4/5] END classifier__max_depth=None, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sam
[CV 5/5] END classifier__max_depth=None, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sam
[CV 1/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sample
[CV 2/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sample
[CV 3/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sample
[CV 4/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sample
[CV 5/5] END classifier__max_depth=5, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sample
[CV 1/5] END classifier__max_depth=None, classifier__max_features=log2, classifier__min_samples_leaf=1, classifier__min_sa
```

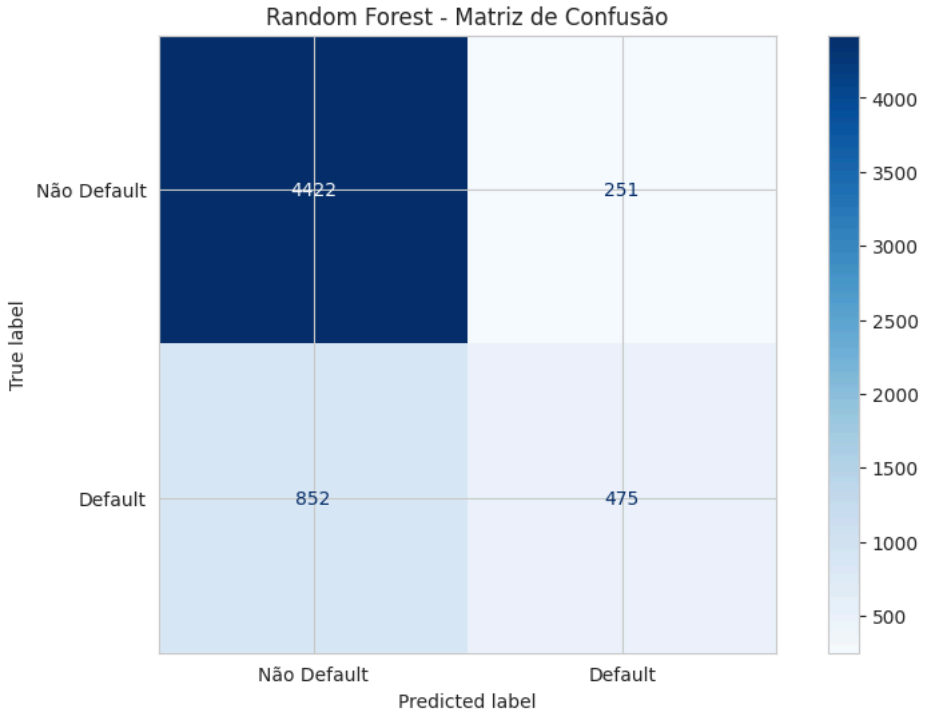
```
[CV 2/5] END classifier__max_depth=15, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sampl
[CV 3/5] END classifier__max_depth=15, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sampl
[CV 4/5] END classifier__max_depth=15, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sampl
[CV 5/5] END classifier__max_depth=15, classifier__max_features=0.5, classifier__min_samples_leaf=2, classifier__min_sampl
[CV 1/5] END classifier__max_depth=20, classifier__max_features=0.5, classifier__min_samples_leaf=5, classifier__min_sampl
[CV 2/5] END classifier__max_depth=20, classifier__max_features=0.5, classifier__min_samples_leaf=5, classifier__min_sampl
[CV 3/5] END classifier__max_depth=20, classifier__max_features=0.5, classifier__min_samples_leaf=5, classifier__min_sampl
[CV 4/5] END classifier__max_depth=20, classifier__max_features=0.5, classifier__min_samples_leaf=5, classifier__min_sampl
[CV 5/5] END classifier__max_depth=20, classifier__max_features=0.5, classifier__min_samples_leaf=5, classifier__min_sampl
[CV 1/5] END classifier__max_depth=20, classifier__max_features=sqrt, classifier__min_samples_leaf=5, classifier__min_samp
[CV 2/5] END classifier__max_depth=20, classifier__max_features=sqrt, classifier__min_samples_leaf=5, classifier__min_samp
[CV 3/5] END classifier__max_depth=20, classifier__max_features=sqrt, classifier__min_samples_leaf=5, classifier__min_samp
[CV 4/5] END classifier__max_depth=20, classifier__max_features=sqrt, classifier__min_samples_leaf=5, classifier__min_samp
[CV 5/5] END classifier__max_depth=20, classifier__max_features=sqrt, classifier__min_samples_leaf=5, classifier__min_samp
[CV 1/5] END classifier__max_depth=20, classifier__max_features=log2, classifier__min_samples_leaf=5, classifier__min_samp
[CV 2/5] END classifier__max_depth=20, classifier__max_features=log2, classifier__min_samples_leaf=5, classifier__min_samp
[CV 3/5] END classifier__max_depth=20, classifier__max_features=log2, classifier__min_samples_leaf=5, classifier__min_samp
[CV 4/5] END classifier__max_depth=20, classifier__max_features=log2, classifier__min_samples_leaf=5, classifier__min_samp
[CV 5/5] END classifier__max_depth=20, classifier__max_features=log2, classifier__min_samples_leaf=5, classifier__min_samp
============================================================
Random Forest otimizado com sucesso!
Tempo total: 1557.5 segundos (26.0 minutos)
Melhor F1-Score (CV): 0.4776
Melhores parâmetros:
   classifier__n_estimators: 100
   classifier__min_samples_split: 2
   classifier__min_samples_leaf: 2
   classifier__max_features: 0.5
   classifier__max_depth: 15
```

```
print('\n' + '='*80)
print('RESULTADOS - RANDOM FOREST')
print('='*80)
print('\n Melhores hiperparâmetros:')
for param, value in random_search_rf.best_params_.items():
    print(f'  {param}: {value}')
print(f'\n Melhor F1-Score médio (CV): {random_search_rf.best_score_:.4f}')

best_rf = random_search_rf.best_estimator_
y_pred_train_rf = best_rf.predict(X_train)
y_pred_test_rf  = best_rf.predict(X_test)
```

```
================================================================================
RESULTADOS - RANDOM FOREST
================================================================================

 Melhores hiperparâmetros:
  classifier__n_estimators: 100
  classifier__min_samples_split: 2
  classifier__min_samples_leaf: 2
  classifier__max_features: 0.5
  classifier__max_depth: 15

 Melhor F1-Score médio (CV): 0.4776
```

```
print(' Avaliando Random Forest...')
print_metrics(y_train, y_pred_train_rf, 'RANDOM FOREST - TREINO')
print_metrics(y_test,  y_pred_test_rf,  'RANDOM FOREST - TESTE')

print('\n=== Classification Report (Teste) ===')
print(classification_report(y_test, y_pred_test_rf,
                            target_names=['Não Default', 'Default'],
                            zero_division=0))

plot_confusion_matrix(y_test, y_pred_test_rf, 'Random Forest - Matriz de Confusão')
```

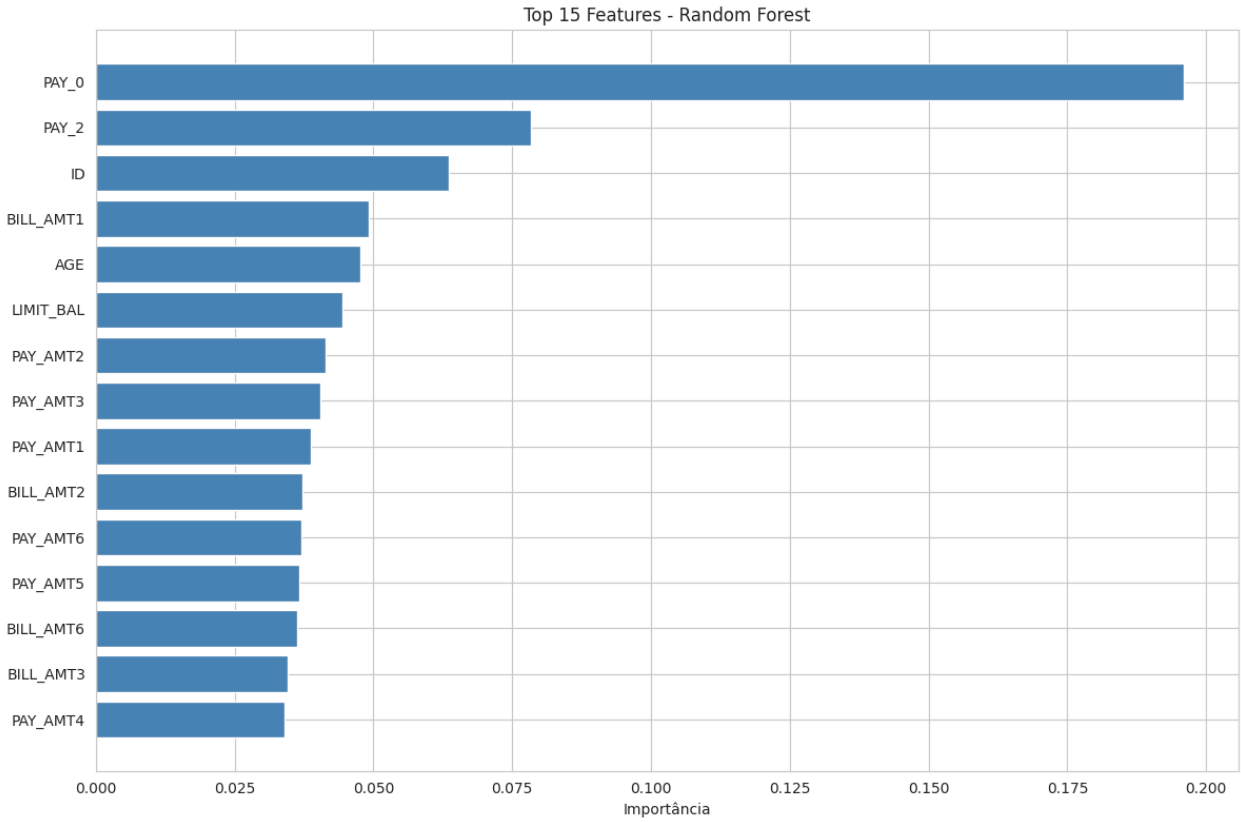
```
 Avaliando Random Forest...

============================================================
                    RANDOM FOREST - TREINO
============================================================
Accuracy : 0.9007
Precision: 0.9513
Recall   : 0.5809
F1-score : 0.7213
============================================================

============================================================
                    RANDOM FOREST - TESTE
============================================================
Accuracy : 0.8162
Precision: 0.6543
Recall   : 0.3580
F1-score : 0.4627
============================================================

=== Classification Report (Teste) ===
              precision    recall  f1-score   support

 Não Default       0.84      0.95      0.89      4673
     Default       0.65      0.36      0.46      1327

    accuracy                           0.82      6000
   macro avg       0.75      0.65      0.68      6000
weighted avg       0.80      0.82      0.79      6000
```




```
print(' Gerando importância das features...')

rf_model    = best_rf.named_steps['classifier']
importances = rf_model.feature_importances_

feature_names_rf = list(numeric_cols)
if categorical_cols:
    cat_enc_rf = best_rf.named_steps['preprocessor'].named_transformers_['categorical'].named_steps['onehotencoder']
    feature_names_rf.extend(cat_enc_rf.get_feature_names_out(categorical_cols))

importance_df = pd.DataFrame({
    'Feature':    feature_names_rf[:len(importances)],
    'Importance': importances
}).sort_values(by='Importance', ascending=False)

print('\nTop 15 Features:')
print(importance_df.head(15).to_string(index=False))

plt.figure(figsize=(12, 8))
top_15 = importance_df.head(15)
plt.barh(top_15['Feature'], top_15['Importance'], color='steelblue')
plt.xlabel('Importância')
plt.title('Top 15 Features - Random Forest')
```

```
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()

print(' Importância das features gerada!')
```

```
 Gerando importância das features...

Top 15 Features:
  Feature  Importance
    PAY_0    0.196029
    PAY_2    0.078379
       ID    0.063610
BILL_AMT1    0.049214
      AGE    0.047582
LIMIT_BAL    0.044461
 PAY_AMT2    0.041489
 PAY_AMT3    0.040511
 PAY_AMT1    0.038685
BILL_AMT2    0.037198
 PAY_AMT6    0.036953
 PAY_AMT5    0.036601
BILL_AMT6    0.036291
BILL_AMT3    0.034649
 PAY_AMT4    0.034005
```




```
 Importância das features gerada!
```

COMPARAÇÃO FINAL DE TODOS OS MODELOS

```
print('📊 Gerando comparação final de todos os modelos...')

final_comparison = pd.DataFrame({
    'Modelo': [
        'Perceptron',      'Perceptron',
        'Árvore Simples',  'Árvore Simples',
        'Árvore Otimizada','Árvore Otimizada',
        'Random Forest',   'Random Forest'
    ],
```